Question 8: How stable is Bitcoin's difficulty adjustment mechanism under large hashrate shocks?

i. Long-run Difficulty trend

Bitcoin's mining difficulty is recalculated every 2016 blocks (one epoch) to maintain a 10-minute block target (Bitref, 2026). A (2022) formalises the retargeting rule as,

$$\delta_{t+1} = \delta_t \times \frac{Actual\ time\ for\ 2016\ blocks}{20160}$$

This expression captures the core stabilising mechanism: if blocks arrive too quickly, the ratio exceeds 1 and difficulty increases; vice versa. The protocol caps any single adjustment at $\pm 300\%$ (4x) (Bitcoin Security Guide, 2026).

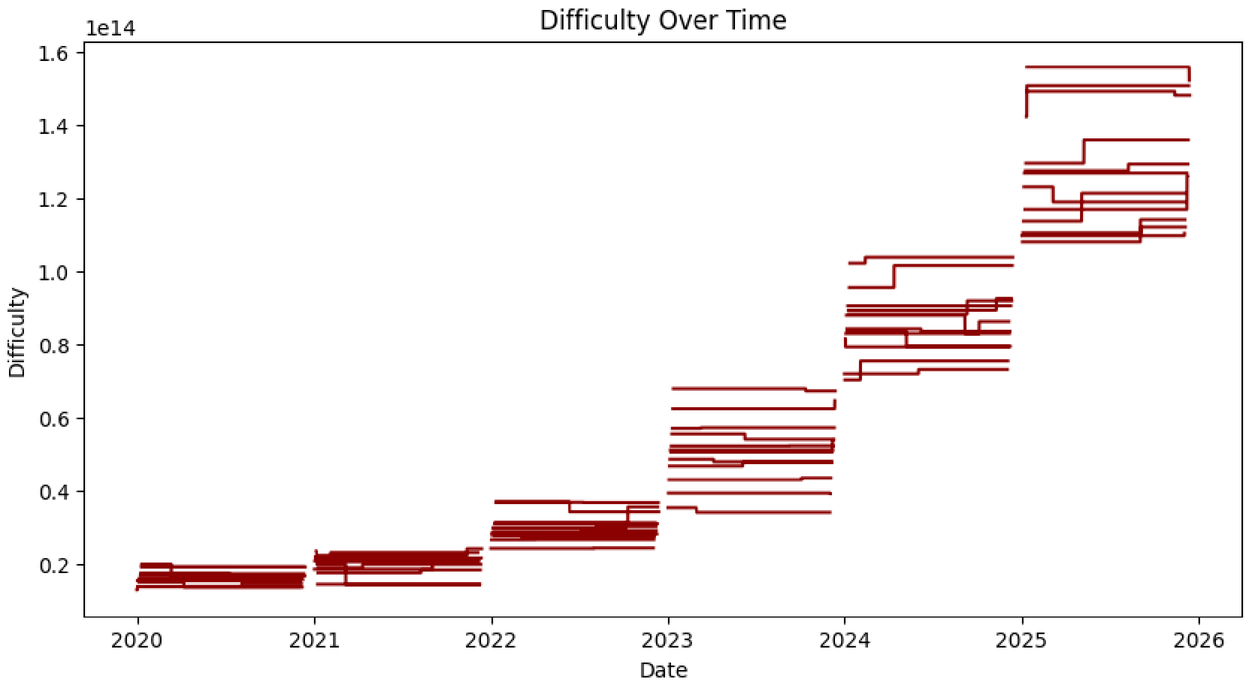

Figure 1: Difficulty increase over time with stepwise difficulty adjustment

Between 2020 and mid-2021, Bitcoin mining difficulty remained relatively stable through a clear disruption that emerged during the China Ban, which removed 65-75% of global mining capacity (Sigalos, 2021). From 2022 to 2023, difficulty rose sharply as the network calibrated and mining activity shifted towards the USA, Russia and Canada (Brownell, 2021). Between 2024 and 2025, difficulty increased exponentially due to advanced ASIC deployment, while institutional miners began dominating through integrated operational efficiencies (Baily et al., 2024; Hashrate index, 2026).

ii. Difficulty Adjustment Resilience under Large Hashrate Shocks

Table 1: Summary statistics of Bitcoin Mining Difficulty and Difficulty Percentage change

Difficulty spread (σ)	***Difficulty mean***	***Difficulty change mean***	***Difficulty change spread***	***Difficulty change (2σ)***	***Difficulty change (3σ)***
$4.11\times \mathbf{10^{13}}$	5.58×10^{13}	0.00086	0.138	0.276	0.414

To identify large hashrate shocks, this analysis defines abnormal events as statistically significant deviations in Bitcoin's difficulty adjustment exceeding (2σ = ±0.27.6%). Block 683424 (May 2021, and 688472.

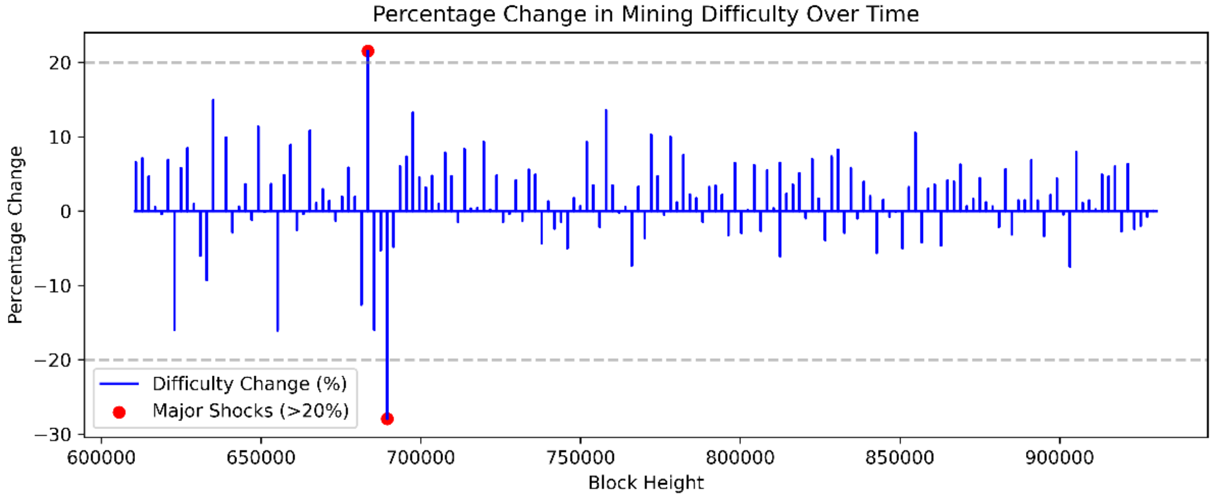

Figure 2: Percentage Change in Mining Difficulty as >20% Major Hash-rate Shock

Table 2: Large Hashrate Events

ID	***Time***	***Difficulty***	***Difficulty Change (%)***
683424	2021-05-13	2.50×10^{13}	21.35
689472	2021-07-03	1.44×10^{13}	-27.63

Podhorsky shows that this adjustment behaves like an "ad valorem" price shock.

$$\frac{\delta_{t+1}}{\delta_t} = 1 + \omega$$

Where, $\omega = \frac{\delta_{t+1}}{\delta_t}$ is the percentage difficulty change. In other words, a difficulty increase acts like a proportional increase in miners' marginal cost, while a difficulty decrease acts like a subsidy.

This farming helps explain why large hash-rate shocks can temporarily destabilise block times. The observed shock of -27.94% on 2021-03-07 is consistent with this model. A drop of that magnitude implies:

$$\omega = -0.2794, \qquad \delta_{t+1} = (1 - 0.2794)\delta_t = 0.7206\delta_t,$$

which is a substantial downward rotation of the supply curve as the new difficulty is 72.60% of the old difficulty. This is a downward adjustment, as the hash rate drops sharply, so blocks are coming in too slowly. The scale of this shock is consistent with China's mining ban. Próspero (2022) defined several reasons for the China Ban. Firstly, regulatory incentives linked to the digital Yuan accelerated the suppression of Bitcoin. Moreover, the computational power required in mining created concerns about regional blackouts, which led to the decommissioning of small hydropower stations to reduce the availability of cheap electricity to miners.

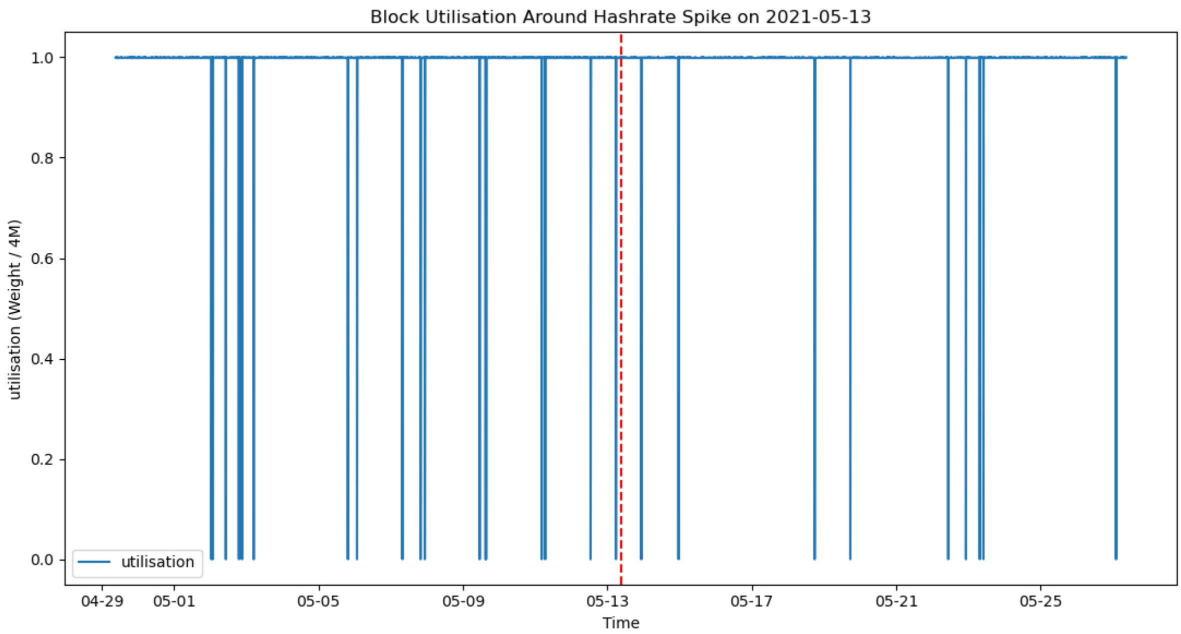

Figure 3: Block Utilisation around Bitcoin Mining ban announcement

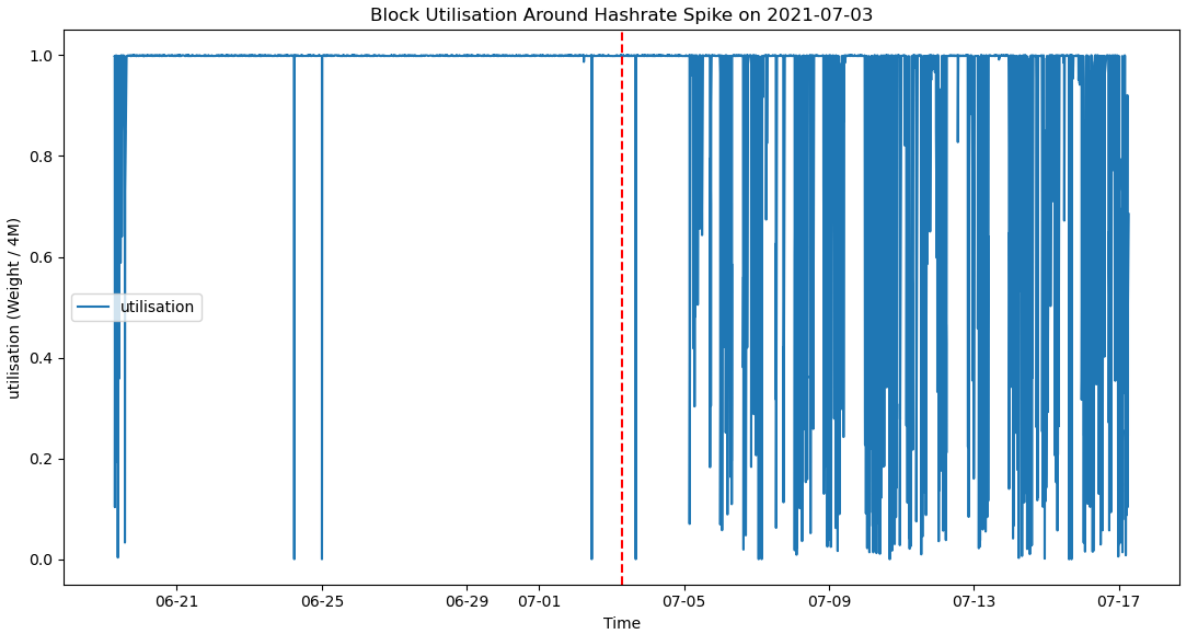

Figure 4: Block Utilisation around the enforcement of Bitcoin Mining Ban

Figure 3-6 shows the utilisation and throughput deterioration. In May 2021, China's State Council announced a crackdown on Bitcoin mining and trading (South China Post, 2021). This announcement caused a mild utilisation dip, but July's full enforcement preceded a severe volatility, frequent low-weight blocks and unstable throughput. The block times between May and July 2021 show persistent volatility as miners exited, relocated and reconnected, as after July 2021, the block time recalibrated towards the target. Additional shocks are defined to test difficulty-adjustment resilience.

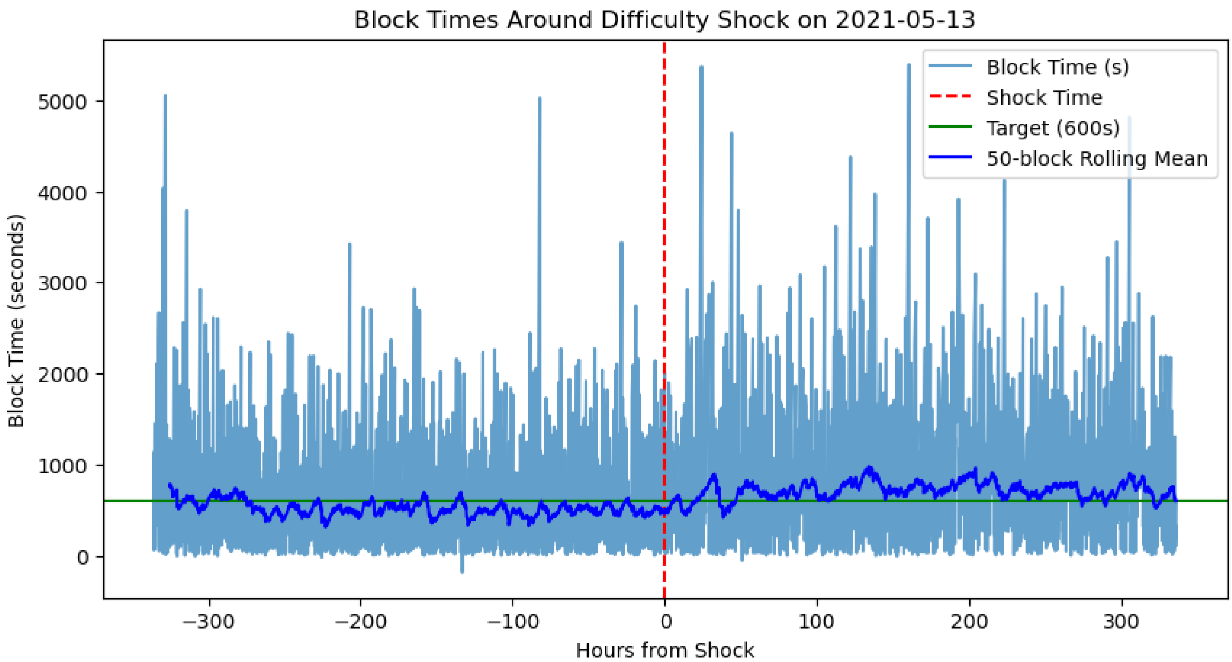

Figure 5: Block Times around the Bitcoin ban announcement, May 2021

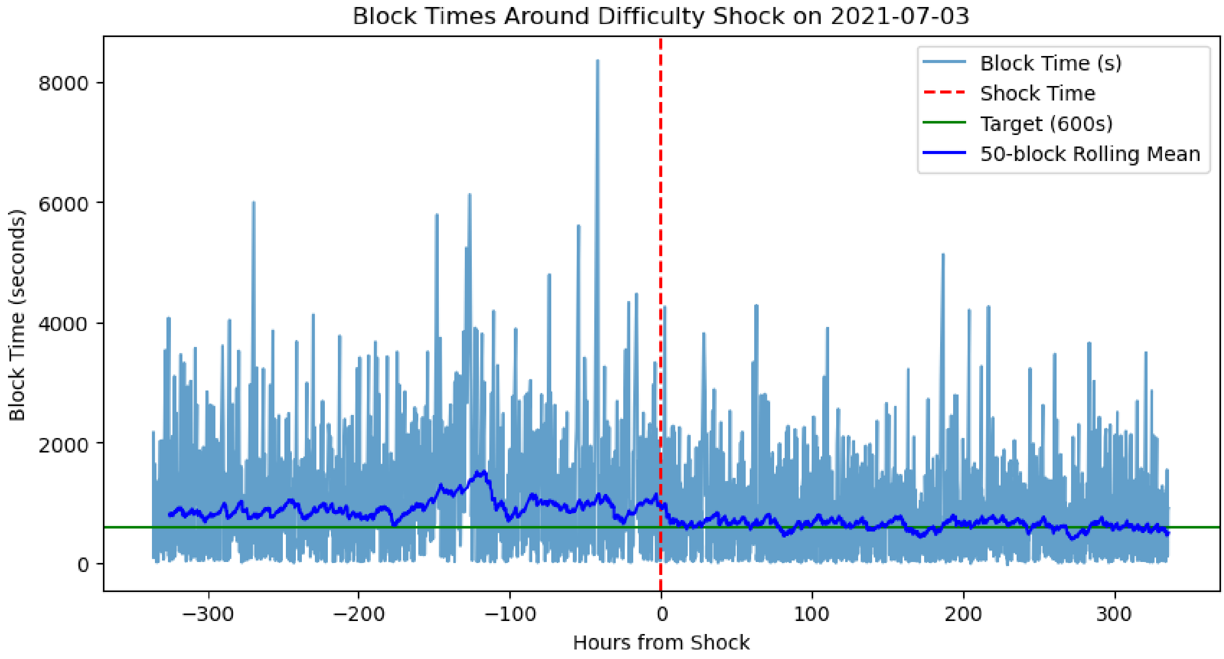

Figure 6: Block Time before and after China Ban Enforcement, July 2021

iii. May 2020 and April 2024 Halving

The Bitcoin halving cuts block rewards by 50%, immediately reducing miner revenue (Meynkhad, 2019). With unchanged operational costs, high-cost miners may exit, briefly lowering the hashrate. This occurred in May 2020, when revenue fell, block times spiked, and difficulty decreased as designed. In contrast, the April 2024 halving caused no block-time disruption, which might be driven by institutional-scale miners with lower marginal cost, explaining why 2020 was more volatile than 2024.

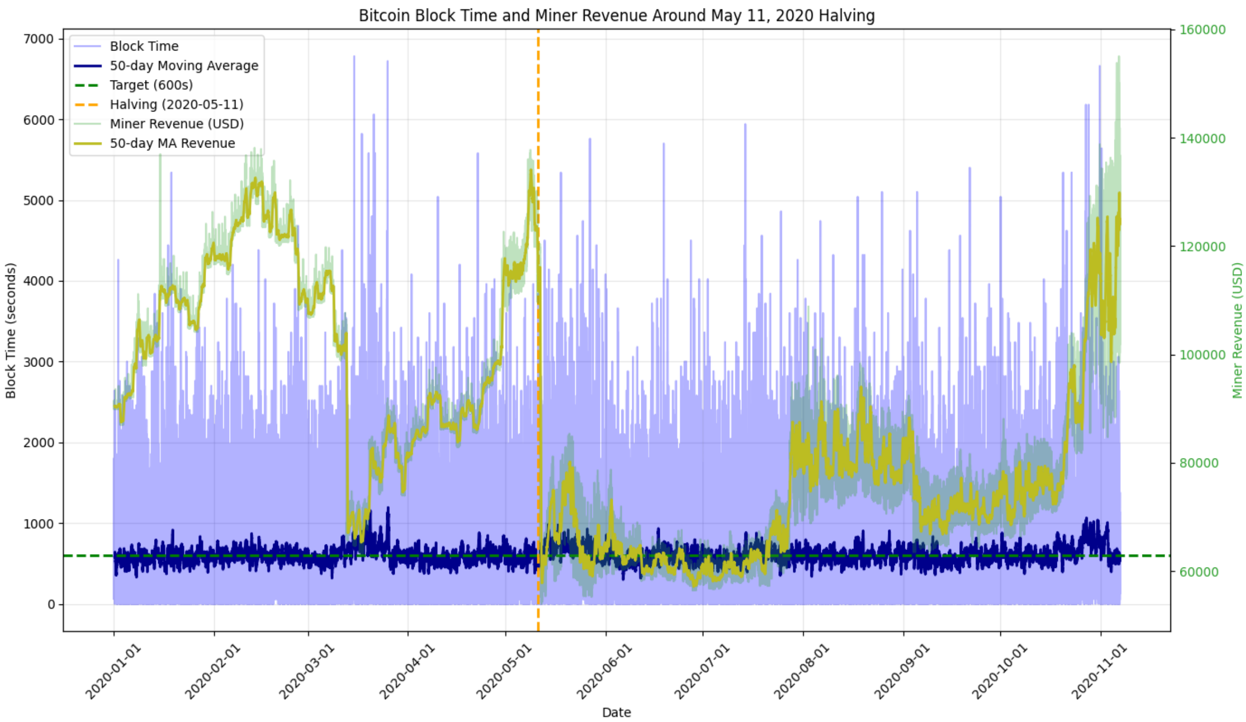

Figure 7: Bitcoin Block Utilisation around 3rd Halving May 11th, 2020

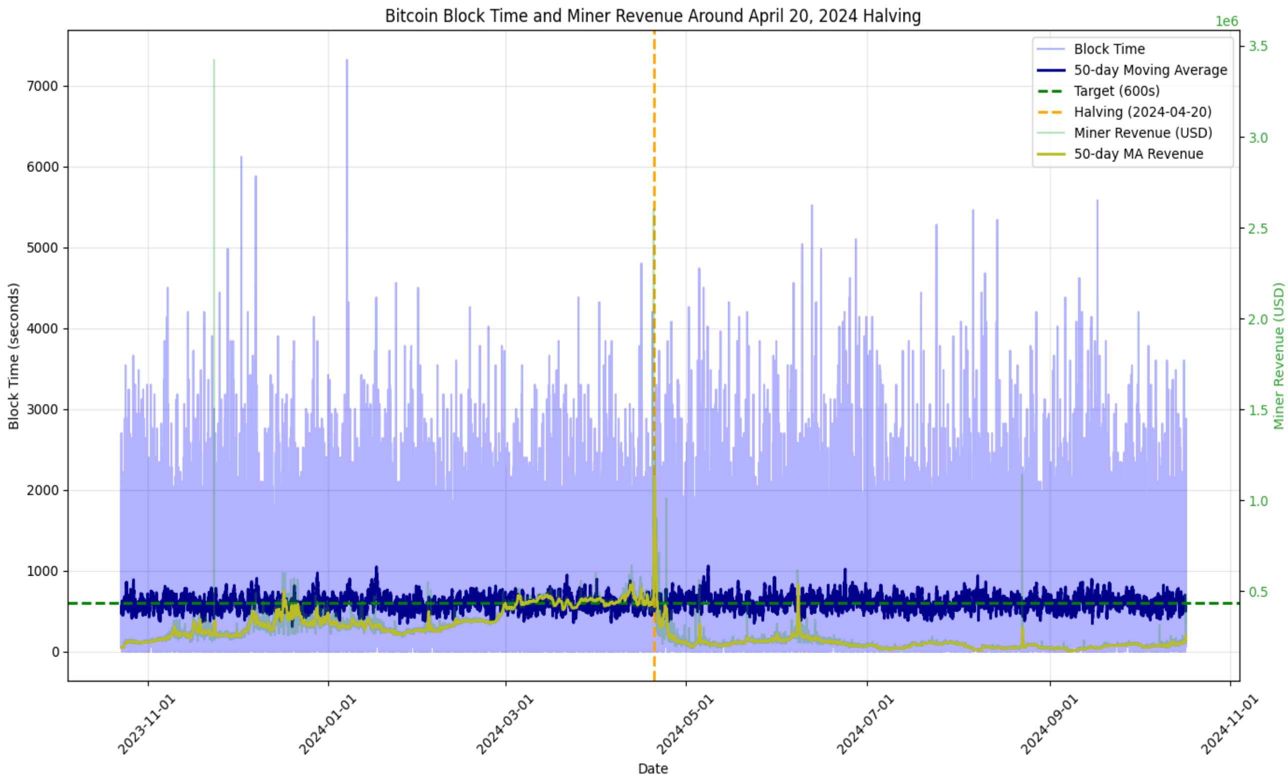

Figure 8: Bitcoin Block Utilisation around 4th Halving April 20, 2024

iv. Chow Test

The Chow test China Ban created the most significant shock, while halving produced a smaller but still statistically significant change.

Table 3: Chow Test

Chow Test for Event Window H_0: Pre-event and Post-event mean block times are equal (no structural break)				
Event	Pre-break mean block time	Post-break mean block time	f-statistic	P-value
3rd Halving May 2020	583.8s	692.4s	29.71	0.000 ***
China Ban Announcement May 2021	492.7s	705.1s	129.84	0.000***
China Ban Enforcement July 2021	812.6s	629.5s	67.27	0.000***
4th Halving April 2024	574.9s	613.7s	4.571	0.0326**

**** significant at 1%, ** significant at 5%*

v. Difficulty Adjustment resilience

The methodology groups blockchain data into epoch intervals and calculates difficulty changes and block-time variability for each period. Firstly, the relationship between shock magnitude (i.e., difficulty change), block time variance, and log block time variance is illustrated using OLS regression. The results show that difficulty shocks have no significant effect on block time variance with explanatory power. ($R^2 \approx 0.01 - 0.02$). This suggests that even large difficulty adjustments (China Ban and Post 2023 miner consolidation era) do not create instability, highlighting the protocol's ability to absorb large hash rate fluctuations. Time-series analysis of the ARIMA(1,0,0) model on lagged shock magnitude shows no persistence in shocks, such as short-lived shocks, which are quickly absorbed by the protocol through periodic difficulty recalibration. However, the GARCH(1,1) model reveals that while shocks themselves are unpredictable, their volatility is highly persistent ($\alpha, \beta = 0.96$). This implies that disruptions such as the 2021 mining ban generate a temporary cluster of elevated uncertainty. meaning volatility emerges in regimes rather than as isolated events.

Table 4: OLS Regressions

OLS regression Linear Model $Block\ time\ variance = \alpha + \beta Shock\ Magnitute$			
Parameter	***Coefficient***	***T-stat***	***P-value***
α	0.00009	2.85	0.005***
β	-0.000006	-1.235	0.219
OLS Regression Multiple Linear Model $\ln(Block\ time\ variance) = \alpha + \beta_o Shock\ Magntute + \beta_1 Post\_Ban + \beta_2 Post\_2023$			
α	12.75	104.8	12.512
β_o	0.4554	0.309	-2.45
β_1	0.0882	0.559	-0.224
β_2	0.0763	0.776	-0.118

Table 5: Autoregressive Model

ARIMA(1,0,0) Model $Shock\ Magnitute = \alpha + \phi_1 Shock\ Magnitute_{t-1} + \varepsilon_1$		
Parameter	Coefficient	P-value
α	0.0447	0.000
ϕ_1	0.0916	0.203
ε_1	0.0018	0.000

Table 6: GARCH(1,1) Model

GARCH(1,1) Model Mean Model : $Shock\ Magnitide_t = \mu + \varepsilon_1$				
Parameter		Coefficient	T-statistics	P-value
μ mean		0.0396	15.2	0.000
Variance Model: $Var_t = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$				
ω	Long-run variance level	0.0001	0.677	4.98
α	Short-run shocks	0.243	0.906	0.365
β	Persistence of volatility	0.715	2.603	0.009

Conclusion

Table 7: Difficulty Adjustments and Recovery Hours of 50-block rolling mean block time returns within ± 60*seconds of the 600-second target*

time	*id*	*Avg. block time pre shock*	*Avg. block time post shock*	*Avg. utilisation pre shock*	*Avg. utilisation post shock*	*Recovery hours*
2021-05-13 08:58:00	683424	514	705	0.99	0.99	8.95
2021-07-03 06:34:00	6894 72	914	629	0.99	0.78	11.45
2020-05-11 00:01:00	629867	597	600	0.83	0.94	10.93
2024-04-20 00:05:00	839997	580	598	0.99	0.99	7.81

Bitcoin's difficulty adjustment mechanism demonstrates strong structural resilience to large hashrate shocks and halving-driven revenue decline. Block times are normally used rapidly, typically within hours, without approaching protocol limits. Even the China mining Ban was absorbed without binding the 4x adjustment cap (0.72<4). This indicates that stability is embedded in the protocol, through short-term deviations reflect a trade-off between responsiveness and long-run equilibrium.